

Ann Mary Thomas

Energy Analyst | Data & Sustainability Analyst | AI/ML Researcher

annmarytttt@gmail.com | linkedin.com/in/ann-mary-thomas-6272aa200 | github.com/Ann-Mary-Thomas | London, UK

★ PUBLISHED · Nature npj Urban Sustainability (July 2026) - TAL-Net: a temporal attention LSTM framework for urban electricity and emissions forecasting in the U.S. | DOI: 10.1038/s42949-026-00424-y

PROFILE

Energy and sustainability analyst combining mechanical engineering foundations with advanced data analytics and deep learning research. Currently at Ameresco, Inc. (NYSE: AMRC) within the Measurement & Verification (M&V) team, delivering data-driven energy performance analysis across clean energy projects in the UK and Europe. Previously embedded at WMG, University of Warwick within 3T Additive Manufacturing as part of the £14.1M government-funded STRATA programme - building energy baselines, designing decarbonisation roadmaps, and delivering Power BI dashboards for Honeywell consortium partners. Research experience at London Metropolitan University's GENESIS Lab, where I co-authored two journal papers (first published in Nature npj Urban Sustainability, July 2026) and designed TAL-Net - a Temporal Attention LSTM Network achieving 0.48% MAPE for electricity demand forecasting. Proven ability to translate complex datasets into clear, actionable insights for technical and non-technical stakeholders across industry, academia, and government bodies.

CORE SKILLS & TECHNOLOGIES

Data Analysis	Python (Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow), SQL, Excel (Advanced), Statistical Analysis
AI & Machine Learning	LSTM, Bidirectional LSTM, Attention Mechanisms (TAL-Net), Time-Series Forecasting, SHAP Interpretability, Hybrid Deep Learning
Visualisation & BI	Power BI, Dashboard Design, KPI Reporting, Data Storytelling, HTML UI Development
Energy & Sustainability	Carbon Footprinting (Scope 1/2/3), CII/IMO Standards, ISO 50001/14001/14040/14044, IPMVP, Load Factor Modelling, ND-NEED 2022, CIBSE Guide F
Systems & Tools	Git, Jupyter Notebook, IoT Sub-metering, Data Pipeline Automation, Data Cleansing & Validation
Professional	Stakeholder Engagement, Technical Report Writing, Grant Reporting, Cross-team Collaboration, Research & Publication

PROFESSIONAL EXPERIENCE

Energy Analyst CURRENT Ameresco, Inc. (NYSE: AMRC) Measurement & Verification (M&V) Team · UK & Europe Clean Energy Projects	Jun 2026 – Present London, UK		
Data Modelling & Analysis • Develop and maintain data-driven energy performance models using utility, metering, and operational datasets to track and evidence energy and carbon savings across client projects • Analyse large, multi-source datasets to identify consumption trends, anomalies, and optimisation opportunities across development, construction, and operations phases • Support development of predictive models and data-driven forecasting approaches for energy and carbon performance			
Automation, Reporting & Stakeholder Communication • Build and enhance automated Python data pipelines to improve delivery efficiency, scalability, and reporting accuracy across M&V workflows • Produce Power BI dashboards, visualisations, and technical reports communicating energy performance insights to technical and non-technical stakeholders • Contribute to M&V plans and performance monitoring strategies aligned to IPMVP methodology			
Energy Consumption & Sustainability Analyst WMG, University of Warwick - 3T Additive Manufacturing (Project STRATA) £14.1M ATI Programme / Innovate UK • Honeywell Consortium • National Centre for Connected Manufacturing	Dec 2025 – Apr 2026		
28,510 kWh Annual facility consumption	6.65 tCO₂ Annual carbon baseline	£9,907/yr Energy cost equivalent	15–18% Projected annual reduction

Energy Baseline & Hotspot Analysis

- Built complete energy performance baseline for a 35-staff facility across **35+ equipment types** - zero sub-meters, zero BMS; applied IPMVP Option A methodology (nameplate ratings, load factor modelling, site observation), validated against ND-NEED 2022, CIBSE Guide F, and Carbon Trust benchmarks (3.26 kWh/employee/day)
- Identified **HVAC (49%), water boiler (25%), and lighting (12%)** as primary consumption hotspots, directing intervention priorities

Scenario Design & Financial Modelling

- Designed **6 improvement scenarios** with 5-year ROI analysis: AC temperature optimisation (18% reduction), automatic shutdown compliance (16%), kitchen automation (59% sub-category) - combined **15–18% annual energy reduction**

Standards, Reporting & Digital Infrastructure

- Structured Energy & Sustainability Policy Framework to **ISO 50001, ISO 14001, ISO 14040/14044**; produced Digital Product Passport (DPP) equipment catalogue to LCA standards
- Specified IoT sub-metering infrastructure (sensor spec, vendor shortlist); participated in ERP/MRP/MES integration discussions for digital twin pipeline
- Built Power BI dashboards (kWh/employee, carbon intensity KPIs); presented findings to WMG management, 3T senior leadership, and NCC/Honeywell consortium partners

Energy Consumption & Sustainability Analyst (Intern)

Sep 2025 – Dec 2025

3T Additive Manufacturing - WMG Industry Placement

- Achieved a **39.3% carbon footprint reduction** through systematic energy data analysis and targeted operational interventions across manufacturing processes
- Constructed facility energy performance baseline from zero - no existing sub-meters, BMS, or monitoring data; applied load factor modelling to nameplate ratings using IPMVP Option A methodology
- Designed and quantified **6 savings scenarios**: AC optimisation (18%), shutdown compliance (16%), kitchen automation (59% sub-category) - combined 15–18% projected annual reduction; produced 5-year financial projections and ROI analysis per scenario
- Aligned recommendations to ISO 50001 (Energy Management Systems); designed sub-metering deployment roadmap to transition from estimated to real-time M&V baseline
- Built interactive Power BI dashboards for Scope 1/2 KPI monitoring; authored technical reports for management and funding bodies - directly leading to full analyst role at WMG

Research Assistant - Energy Forecasting & Carbon Analytics

Feb 2025 – Present

GENESIS Research Lab, London Metropolitan University (under Dr. Maitreyee Dey)

0.48% MAPE - electricity demand	1.60% MAPE - CO ₂ emissions	19× Improvement over ARIMA	p < 0.0001 Statistical significance
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Model Architecture & Results

- **Co-authored 2 peer-reviewed journal papers** on electricity demand and CO₂ emissions forecasting - first paper published in Nature npj Urban Sustainability (July 2026); second under review
- Designed **TAL-Net (Temporal Attention LSTM Network)** - bidirectional LSTM + temporal attention mechanism; trained on hourly US EIA grid data (California & Texas); achieved **0.48% MAPE (demand)** and **1.60% MAPE (CO₂)**; outperformed 4 baselines (RNN, LSTM, RBM+MLP, RBM+LSTM); **19× improvement over ARIMA**

Feature Engineering, Interpretability & Scenario Analysis

- Engineered 27 input features (temporal indicators, lag variables 1h/24h/168h, rolling stats, fuel mix, weather); VIF analysis reduced 28→24 features without accuracy loss
- SHAP analysis confirmed coal (68.38%) and natural gas (23.99%) as dominant CO₂ drivers - physically consistent and audit-ready
- Solar penetration scenarios (10–50%): **50% penetration achieved 6.74% CO₂ reduction** (~127,000 car-equivalents annually); validated with paired t-tests, Bonferroni correction, bootstrap CI, Cohen's d - all **p < 0.0001**

Junior Data Analyst - Energy & Emissions Performance

Aug 2022 – Jan 2024

Navalt Solar and Electric Boats Pvt. Ltd

700+ Vessels processed	200,000+ Rows / 200+ metrics	85% Efficiency gain (1 wk → 4 hrs)	4 docs Condensed into CII Guide
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- Built **automated Python ETL pipeline** (pandas, NumPy) processing 200,000+ rows across 700+ vessels (200+ metrics) - weekly data reporting reduced from ~1 week to ~4 hours (**85% efficiency gain**)
- Performed **CII (Carbon Intensity Indicator)** calculations per IMO regulation (A–E rating: CO₂ ÷ DWT × distance) for full fleet; calculated SFOC/SFOS metrics for engine health monitoring and vessel performance benchmarking

- Prepared quarterly ESI (Environmental Ship Index) compliance reports for port authorities; maintained sea trial, drydock, and vessel performance databases
- Condensed 4 dense IMO regulatory documents into a single CII Reference Guide; built predictive models for fuel efficiency and CO₂ emissions using engine performance variables

RESEARCH & PUBLICATIONS

- **[PUBLISHED] Nature npj Urban Sustainability (July 2026)** - TAL-Net: a temporal attention LSTM framework for urban electricity and emissions forecasting in the U.S. - Ann Mary Thomas, Maitreyee Dey & Soumya Prakash Rana. DOI: 10.1038/s42949-026-00424-y | Open Access
- **Second peer-reviewed journal paper** co-authored - AI-based CO₂ emissions forecasting using TAL-Net; currently under review
- Energy demand & CO₂ forecasting: **0.48% MAPE (demand), 1.60% MAPE (CO₂), 19× improvement over ARIMA**; scenario modelling shows 6.74% CO₂ reduction at 50% solar penetration (~127,000 car-equivalents removed annually)
- Student volunteer - FICTA'25 (Frontiers of Intelligent Computing: Theory and Applications), international conference

SELECTED PROJECTS

Vehicle Emissions Power BI Dashboard - Interactive dashboard on CO₂ emissions across vehicle types (EU dataset); identified engine size as ~70% of emission variance with multi-dimensional manufacturer/country/fuel-type filters.

Graduate Outcomes Analysis (Python) - Random Forest classification and K-means clustering on UK HESA data; identified four graduate profiles for targeted university support recommendations.

Marketing Campaign Response Model (Python) - Logistic regression on 1,500 customer records; identified demographic features improving targeting accuracy by 20%.

EDUCATION

MSc in Data Analytics - London Metropolitan University	2024 – 2025
B.Tech in Mechanical Automobile Engineering - SCT College of Engineering	2017 – 2021

KEY ACHIEVEMENTS

- **Published in Nature npj Urban Sustainability (July 2026)** - First author on TAL-Net paper; DOI: 10.1038/s42949-026-00424-y
- **39.3% carbon footprint reduction** at 3T Additive Manufacturing through data-driven sustainability analysis and targeted decarbonisation interventions
- **Co-authored 2 peer-reviewed papers** - TAL-Net achieved 0.48% MAPE (demand), 1.60% MAPE (CO₂), 19× improvement over ARIMA (p < 0.0001)
- **£14.1M programme contribution** - delivered audited energy baseline (28,510 kWh / 6.65 tCO₂) for Innovate UK STRATA, validated against ND-NEED 2022 and CIBSE Guide F
- **85% efficiency gain** at Navalt - automated Python pipeline cut weekly data reporting from 1 week to 4 hours across 700+ vessels and 200,000+ data rows
- Progressed directly from intern to full WMG analyst role following performance at 3T Additive Manufacturing

CERTIFICATIONS

- Excel for Data Analytics - Simplilearn
- Statistics for Data Science - Simplilearn